

Multi Frame ROI Mode

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

The Multi Frame function on the ROI (Region Of Interest) mode is described in this document.

This function is became possible the sequence control by setting the gain and the exposure period of each frame, assuming Multi Frame ROI 2 frames or Multi Frame ROI 4 frames to be one-set.

Cropping position can set maximum 16 areas that specified by horizontal 4 points and vertical 4 points, regarding effective pixel start position as origin (0, 0) in all pixel scan mode.

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3. Features

◆ Multi Frame ROI mode

2 frame set (Multi Frame ROI mode 2 frame)

4 frame set (Multi Frame ROI mode 4 frame)

◆ Frame identifier

Frame information line (In case of SLVS output)

Embedded data line (In case of SLVS – EC output or CSI-2 serial data output)

◆ Support mode

Normal mode (XVS / XHS sync)

* Trigger mode is not supported.

◆ Support readout drive mode

All-pixel scan mode

◆ Support ROI mode

ROI mode

Overlap ROI mode

◆ ROI function

Cropping position

ROI mode : Maximum 16 areas (horizontal 4 points and vertical 4 points)

Overlap ROI mode : Maximum 4 areas

Multi Frame ROI Mode is able to set the cropping position on each frame.

◆ The monitor output for Exposure period

◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

◆ Support each frame gain adjust function

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4. Register Map

The register map for Multi Frame ROI mode is shown below.

Please refer to the product specification for register setup other than those list.

Registers corresponding to Chip ID = 03h in Write mode. (I²C:31**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
00h	3100h	0	ROI_MODE [0]	ROI movement mode change 0: ROI Mode 1: Overlap ROI Mode	0	00h	I
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
04h	3104h	0	FID0_ROIH1ON [0]	The horizontal setting of FID0 ROI area (1, y) (y = 1 to 4) 0: Disable 1: Enable	0	00h	V
		1	FID0_ROIV1ON [0]	The vertical setting of FID0 ROI area (x, 1) (x = 1 to 4) 0: Disable 1: Enable	0		I
		2	FID0_ROIH2ON [0]	The horizontal setting of FID0 ROI area (2, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		3	FID0_ROIV2ON [0]	The vertical setting of FID0 ROI area (x, 2) (x = 1 to 4) 0: Disable 1: Enable	0		I
		4	FID0_ROIH3ON [0]	The horizontal setting of FID0 ROI area (3, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		5	FID0_ROIV3ON [0]	The vertical setting of FID0 ROI area (x, 3) (x = 1 to 4) 0: Disable 1: Enable	0		I
		6	FID0_ROIH4ON [0]	The horizontal setting of FID0 ROI area (4, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		7	FID0_ROIV4ON [0]	The vertical setting of FID0 ROI area (x, 4) (x = 1 to 4) 0: Disable 1: Enable	0		I
06h	3106h	0	FID1_ROIH1ON [0]	The horizontal setting of FID1 ROI area (1, y) (y = 1 to 4) 0: Disable 1: Enable	0	00h	V
		1	FID1_ROIV1ON [0]	The vertical setting of FID1 ROI area (x, 1) (x = 1 to 4) 0: Disable 1: Enable	0		I
		2	FID1_ROIH2ON [0]	The horizontal setting of FID1 ROI area (2, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		3	FID1_ROIV2ON [0]	The vertical setting of FID1 ROI area (x, 2) (x = 1 to 4) 0: Disable 1: Enable	0		I
		4	FID1_ROIH3ON [0]	The horizontal setting of FID1 ROI area (3, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		5	FID1_ROIV3ON [0]	The vertical setting of FID1 ROI area (x, 3) (x = 1 to 4) 0: Disable 1: Enable	0		I
		6	FID1_ROIH4ON [0]	The horizontal setting of FID1 ROI area (4, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		7	FID1_ROIV4ON [0]	The vertical setting of FID1 ROI area (x, 4) (x = 1 to 4) 0: Disable 1: Enable	0		I

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
08h	3108h	0	FID2_ROIH1ON [0]	The horizontal setting of FID2 ROI area (1, y) (y = 1 to 4) 0: Disable 1: Enable	0	00h	V
		1	FID2_ROIV1ON [0]	The vertical setting of FID2 ROI area (x, 1) (x = 1 to 4) 0: Disable 1: Enable	0		I
		2	FID2_ROIH2ON [0]	The horizontal setting of FID2 ROI area (2, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		3	FID2_ROIV2ON [0]	The vertical setting of FID2 ROI area (x, 2) (x = 1 to 4) 0: Disable 1: Enable	0		I
		4	FID2_ROIH3ON [0]	The horizontal setting of FID2 ROI area (3, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		5	FID2_ROIV3ON [0]	The vertical setting of FID2 ROI area (x, 3) (x = 1 to 4) 0: Disable 1: Enable	0		I
		6	FID2_ROIH4ON [0]	The horizontal setting of FID2 ROI area (4, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		7	FID2_ROIV4ON [0]	The vertical setting of FID2 ROI area (x, 4) (x = 1 to 4) 0: Disable 1: Enable	0		I
0Ah	310Ah	0	FID3_ROIH1ON [0]	The horizontal setting of FID3 ROI area (1, y) (y = 1 to 4) 0: Disable 1: Enable	0	00h	V
		1	FID3_ROIV1ON [0]	The vertical setting of FID3 ROI area (x, 1) (x = 1 to 4) 0: Disable 1: Enable	0		I
		2	FID3_ROIH2ON [0]	The horizontal setting of FID3 ROI area (2, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		3	FID3_ROIV2ON [0]	The vertical setting of FID3 ROI area (x, 2) (x = 1 to 4) 0: Disable 1: Enable	0		I
		4	FID3_ROIH3ON [0]	The horizontal setting of FID3 ROI area (3, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		5	FID3_ROIV3ON [0]	The vertical setting of FID3 ROI area (x, 3) (x = 1 to 4) 0: Disable 1: Enable	0		I
		6	FID3_ROIH4ON [0]	The horizontal setting of FID3 ROI area (4, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		7	FID3_ROIV4ON [0]	The vertical setting of FID3 ROI area (x, 4) (x = 1 to 4) 0: Disable 1: Enable	0		I

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
20h	3120h	0	FID0_ROIPH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID0 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
21h	3121h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
22h	3122h	0	FID0_ROIPV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID0 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
23h	3123h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
24h	3124h	0	FID0_ROIWH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID0 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
25h	3125h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
26h	3126h	0	FID0_ROIWW1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
27h	3127h	7		Designation of vertical cropping size for FID0 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
28h	3128h	7		Fixed to 0			
		0	FID0_ROIPH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID0 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
29h	3129h	7		MSB			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
2Ah	312Ah	7		Fixed to 0			
		0	FID0_ROIPV2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID0 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
2Bh	312Bh	7		MSB			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
2Bh	312Bh	7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
2Ch	312Ch	0	FID0_ROIWH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID0 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
2Dh	312Dh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
2Eh	312Eh	0	FID0_ROIWH2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID0 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
2Fh	312Fh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
30h	3130h	0	FID0_ROIPH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID0 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
31h	3131h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
32h	3132h	0	FID0_ROIPV3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID0 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
33h	3133h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
34h	3134h	0	FID0_ROIWH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID0 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
35h	3135h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
36h	3136h	0	FID0_ROIWW3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID0 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
37h	3137h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
38h	3138h	0	FID0_ROIPH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID0 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
39h	3139h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
3Ah	313Ah	0	FID0_ROIPV4 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
3Bh	313Bh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
3Ch	313Ch	0	FID0_ROIWH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID0 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
3Dh	313Dh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
3Eh	313Eh	0	FID0_ROIWW4 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
3Fh	313Fh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
60h	3160h	0	FID1_ROIPV1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID1 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
61h	3161h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
62h	3162h	0	FID1_ROIPV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID1 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
63h	3163h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
64h	3164h	0	FID1_ROIWH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID1 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
65h	3165h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
66h	3166h	0	FID1_ROIWH1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID1 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
67h	3167h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
68h	3168h	0	FID1_ROIPH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID1 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
69h	3169h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
6Ah	316Ah	0	FID1_ROIPV2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
6Bh	316Bh	7	FID1_ROIPV2 [12:0]	Designation of vertical cropping position for FID1 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}	0000h	00h	I
		0					
		1					
		2					
		3					
		4					
		5					
6Bh	316Bh	6	FID1_ROIPV2 [12:0]	MSB	0000h	00h	—
		7					
		Fixed to 0					
		Fixed to 0					
6Ch	316Ch	0	FID1_ROIWH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
6Dh	316Dh	7	FID1_ROIWH2 [12:0]	Designation of horizontal cropping size for FID1 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
6Dh	316Dh	6	FID1_ROIWH2 [12:0]	MSB	0000h	00h	—
		7					
		Fixed to 0					
		Fixed to 0					
6Eh	316Eh	0	FID1_ROIWH2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
6Fh	316Fh	7	FID1_ROIWH2 [12:0]	Designation of vertical cropping size for FID1 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}	0000h	00h	I
		0					
		1					
		2					
		3					
		4					
		5					
6Fh	316Fh	6	FID1_ROIWH2 [12:0]	MSB	0000h	00h	—
		7					
		Fixed to 0					
		Fixed to 0					
6Fh	316Fh	0	FID1_ROIWH2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
6Fh	316Fh	7	FID1_ROIWH2 [12:0]	MSB	0000h	00h	—
		0					
		Fixed to 0					
		Fixed to 0					

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
70h	3170h	0	FID1_ROIPH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID1 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
71h	3171h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
72h	3172h	0	FID1_ROIPV3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID1 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
73h	3173h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
74h	3174h	0	FID1_ROIWH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID1 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
75h	3175h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
76h	3176h	0	FID1_ROI WV3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID1 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
77h	3177h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
78h	3178h	0	FID1_ROI PH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID1 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
79h	3179h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
7Ah	317Ah	0	FID1_ROI PV4 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID1 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
7Bh	317Bh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
7Ch	317Ch	0	FID1_ROIWH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
7Dh	317Dh	7		Designation of horizontal cropping size for FID1 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
7Eh	317Eh	7		Fixed to 0			
		0	FID1_ROIWH4 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
7Fh	317Fh	7		Designation of vertical cropping size for FID1 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
80h	3180h	7		Fixed to 0			
		0	FID1_ROIWH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
81h	3181h	7		Designation of horizontal cropping position for FID2 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
82h	3182h	0	FID2_ROIPV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID2 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
83h	3183h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
84h	3184h	0	FID2_ROIWH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID2 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
85h	3185h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
86h	3186h	0	FID2_ROI WV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID2 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
87h	3187h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
88h	3188h	0	FID2_ROIPH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID2 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
89h	3189h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
8Ah	318Ah	0	FID2_ROIPV2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID2 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
8Bh	318Bh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
8Ch	318Ch	0	FID2_ROIWH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID2 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
8Dh	318Dh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
8Eh	318Eh	0	FID2_ROIWW2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID2 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
8Fh	318Fh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
90h	3190h	0	FID2_ROIPH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID2 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
91h	3191h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
92h	3192h	0	FID2_ROIPV3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID2 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
93h	3193h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
94h	3194h	0	FID2_ROIWH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
95h	3195h	7		Designation of horizontal cropping size for FID2 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID2_ROIWH3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0	0		—
96h	3196h	7		Fixed to 0	0		—
		0	FID2_ROIWH3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
97h	3197h	7		Designation of vertical cropping size for FID2 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID2_ROIWH3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0	0		—
98h	3198h	7		Fixed to 0	0		—
		0	FID2_ROIWH3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
99h	3199h	7		Designation of horizontal cropping position for FID2 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID2_ROIWH3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0	0		—
99h	3199h	7		Fixed to 0	0		—
		0	FID2_ROIWH3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0	0		—
99h	3199h	7		Fixed to 0	0		—

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
9Ah	319Ah	0	FID2_ROIPV4 [12:0]	LSB	0000h	00h	I
		1		Designation of vertical cropping position for FID2 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		2					
		3					
		4					
		5					
		6					
7							
9Bh	319Bh	0		MSB	0	00h	—
		1					
		2					
		3					
		4					
		5	Fixed to 0				
		6	Fixed to 0				
7	Fixed to 0						
9Ch	319Ch	0	FID2_ROIWH4 [12:0]	LSB	0000h	00h	V
		1		Designation of horizontal cropping size for FID2 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		2					
		3					
		4					
		5					
		6					
7							
9Dh	319Dh	0		MSB	0	00h	—
		1					
		2					
		3					
		4					
		5	Fixed to 0				
		6	Fixed to 0				
7	Fixed to 0						
9Eh	319Eh	0	FID2_ROI WV4 [12:0]	LSB	0000h	00h	I
		1		Designation of vertical cropping size for FID2 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		2					
		3					
		4					
		5					
		6					
7							
9Fh	319Fh	0		MSB	0	00h	—
		1					
		2					
		3					
		4					
		5	Fixed to 0				
		6	Fixed to 0				
7	Fixed to 0						

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
A0h	31A0h	0	FID3_ROIPH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID3 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
A1h	31A1h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
A2h	31A2h	0	FID3_ROIPV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID3 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
A3h	31A3h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
A4h	31A4h	0	FID3_ROIWH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID3 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
A5h	31A5h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
A6h	31A6h	0	FID3_ROI WV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID3 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
A7h	31A7h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
A8h	31A8h	0	FID3_ROI PH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID3 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
A9h	31A9h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
AAh	31AAh	0	FID3_ROI PV2 [11:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID3 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
ABh	31ABh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
ACh	31ACh	0	FID3_ROIWH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID3 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
ADh	31ADh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
AEh	31AEh	0	FID3_ROIWH2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID3 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
AFh	31AFh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
B0h	31B0h	0	FID3_ROIPH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID3 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
B1h	31B1h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
B2h	31B2h	0	FID3_ROIPV3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID3 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
B3h	31B3h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
B4h	31B4h	0	FID3_ROIWH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID3 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
B5h	31B5h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
B6h	31B6h	0	FID3_ROIWW3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size for FID3 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
B7h	31B7h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
B8h	31B8h	0	FID3_ROIPH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping position for FID3 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
B9h	31B9h	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
BAh	31BAh	0	FID3_ROIPV4 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping position for FID3 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
BBh	31BBh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
BCh	31BCh	0	FID3_ROIWH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size for FID3 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		7					
BDh	31BDh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing	
4-wire	I ² C				By register	By address		
BEh	31BEh	0	FID3_ROIWV4 [12:0]	LSB	0000h	00h	I	
		1		Designation of vertical cropping size for FID3 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}				
		2						
		3						
		4						
		5						
		6						
7								
BFh	31BFh	0		MSB	0	00h		—
		1						
		2						
		3						
		4						
		5						
		6						
7		Fixed to 0	0	—				
		Fixed to 0	0	—				
		Fixed to 0	0	—				

^{*1} IMX530 series and IMX535 series.

^{*2} IMX540 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

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Registers corresponding to Chip ID = 04h in Write mode. (I²C:32**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
08h	3208h	0	WDSEL [1:0]	Multi frame number setting 0d: Normal mode, 1d: Multi Frame ROI mode 2 frames 2d: Setting prohibited 3d: Multi Frame ROI mode 4 frames	0h	00h	V
		1					
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
40h	3240h	0		LSB		48h*1	70h*2
		1					
		2					
		3					
		4					
		5					
		6					
		7					
41h	3241h	0	SHS1 [23:0]	Storage time adjustment Designated in line unit In Multi Frame ROI mode, designated for FID: 1. (2 Frame and 4 Frame commonness)	000048h*1	000070h*2	00h
		1					
		2					
		3					
		4					
		5					
		6					
		7					
42h	3242h	0		MSB			00h
		1					
		2					
		3					
		4					
		5					
		6					
		7					

Address		bit	Register Name	Description	Default value after reset				Reflection timing								
4-wire	I ² C				By register		By address										
44h	3244h	0	SHS2 [23:0]	LSB	000048h ^{*1}	000070h ^{*2}	48h ^{*1}	70h ^{*2}	V								
		1															
		2															
		3															
		4															
		5															
		6															
		7															
45h	3245h	0		Storage time adjustment Designated in line unit			In Multi Frame ROI mode 2 frame, designated for FID: 0. In Multi Frame ROI mode 4 frame, designated for FID: 2.	000048h ^{*1}		000070h ^{*2}	00h	V					
		1															
		2															
		3															
		4															
		5															
		6															
		7															
46h	3246h	0											000048h ^{*1}	000070h ^{*2}	00h	V	
		1															
		2															
		3															
		4															
		5															
		6															
		7															
		0	MSB		000048h ^{*1}	000070h ^{*2}									V		
		1															
		2															
		3															
		4															
		5															
		6															
		7															
48h	3248h	0	SHS3 [23:0]				LSB	000048h ^{*1}	000070h ^{*2}	48h ^{*1}		70h ^{*2}					V
		1															
		2															
		3															
		4															
		5															
		6															
		7															
49h	3249h	0					Storage time adjustment Designated in line unit			In Multi Frame ROI mode 4 frame, designated for FID: 3	000048h ^{*1}	000070h ^{*2}	00h	V			
		1															
		2															
		3															
		4															
		5															
		6															
		7															
4Ah	324Ah	0				000048h ^{*1}	000070h ^{*2}						00h		V		
		1															
		2															
		3															
		4															
		5															
		6															
		7															

Address		bit	Register Name	Description	Default value after reset				Reflection timing								
4-wire	I ² C				By register		By address										
4Ch	324Ch	0	4 [23:0]	LSB	000048h ^{*1}	000070h ^{*2}	48h ^{*1}	70h ^{*2}	V								
		1															
		2															
		3															
		4															
		5															
		6															
		7															
4Dh	324Dh	0		Storage time adjustment Designated in line unit			In Multi Frame ROI mode 4 frame, designated for FID: 0	000048h ^{*1}		000070h ^{*2}	00h		V				
		1															
		2															
		3															
		4															
		5															
		6															
		7															
4Eh	324Eh	0									In Multi Frame ROI mode 4 frame, designated for FID: 0	000048h ^{*1}		000070h ^{*2}	00h		V
		1															
		2															
		3															
		4															
		5															
		6															
		7															
				MSB													

^{*1} IMX530 series and IMX540 series.

^{*2} IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

Registers corresponding to Chip ID = 06h in Write mode. (I²C:34**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
36h	3436h	0	TOUT0SEL [2:0]	TOUT0 pin setting	0h	00h	S
		1		0h : Low fixed			
		2		1h : Exposure period monitoring			
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—

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Registers corresponding to Chip ID = 07h in Write mode. (I²C:35**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
00h	3500h	0		Fixed to 1	1	01h	—
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6	WD_GAIN_MODE [0]	Setting of Multi Frame ROI gain function 0 : Normal mode 1 : Each frame gain mode	0		S
		7		Fixed to 0	0	—	
14h	3514h	0	GAIN [8:0]	LSB	000h	00h	V
		1		Setting of gain adjustment Effective at "WD_GAIN_MODE = 0" Designated for FID: 0 to 3 identical gain			
		2					
		3					
		4		Effective at "WD_GAIN_MODE = 1" Designated for FID: 0			
		5					
		6					
		7					
15h	3515h	0		MSB	0	00h	—
		1					—
		2					—
		3					—
		4					—
		5					—
		6					—
		7					—
16h	3516h	0	GAIN1 [8:0]	LSB	000h	00h	V
		1		Setting of gain adjustment Effective at "WD_GAIN_MODE = 1" Designated for FID: 1			
		2					
		3					
		4					
		5					
		6					
		7					
17h	3517h	0		MSB	0	00h	—
		1					—
		2					—
		3					—
		4					—
		5					—
		6					—
		7					—

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
18h	3518h	0	GAIN2 [8:0]	LSB	000h	00h	V
		1					
		2					
		3					
		4		Setting of gain adjustment Effective at "WD_GAIN_MODE = 1" Designated for FID: 2			
		5					
		6					
		7					
19h	3519h	0		MSB	0	00h	—
		1					
		2					
		3					
		4					
		5					
		6					
		7					
1Ah	351Ah	0	GAIN3 [8:0]	LSB	000h	00h	V
		1					
		2					
		3					
		4		Setting of gain adjustment Effective at "WD_GAIN_MODE = 1" Designated for FID: 3			
		5					
		6					
		7					
1Bh	351Bh	0		MSB	0	00h	—
		1					
		2					
		3					
		4					
		5					
		6					
		7					

5. Setting concerning Multi Frame ROI

5.1. In Case of ROI

The setting of ROI is basically similar to the datasheet. Refer to the datasheet for details.

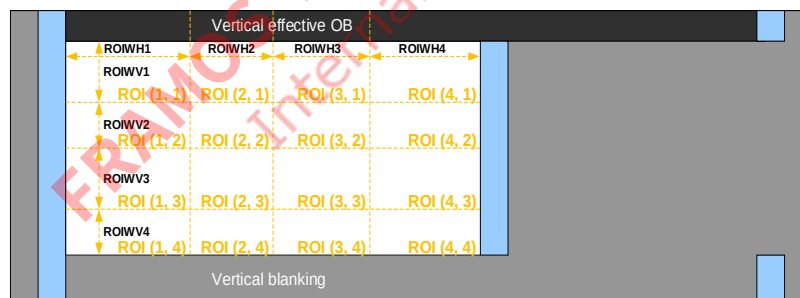
The setting of Multi Frame concerns is basically similar to the application note. Refer to the application note of Multi Frame Set Output mode.

However, Multi Frame ROI have the following restrictions.

- ◆ Cropping position can set maximum 16 areas that specified by horizontal 4 points and vertical 4 points, regarding effective pixel start position as origin (0, 0) in all pixel scan mode.



Image Drawing of Designated Areas in ROI Mode (SLVS)



Details of Image Drawing (SLVS)

- ◆ Multi Frame ROI is able to set cropping address of each frame. But set the same values of ROI cropping size of each frame.

Set the same values to the following registers.

Multi Frame ROI 2 frames:

Designation of horizontal cropping size: FID0_ROIWH1 = FID1_ROIWH1, ...FID0_ROIWH4 = FID1_ROIWH4

Designation of vertical cropping size: FID0_ROIWH1 = FID1_ROIWH1, ...FID0_ROIWH4 = FID1_ROIWH4

Multi Frame ROI 4 frames:

Designation of horizontal cropping size: FID0_ROIWH1 = FID1_ROIWH1 = FID2_ROIWH1 = FID3_ROIWH1

⋮
FID0_ROIWH4 = FID1_ROIWH4 = FID2_ROIWH4 = FID3_ROIWH4

Designation of vertical cropping size: FID0_ROIWH1 = FID1_ROIWH1 = FID2_ROIWH1 = FID3_ROIWH1

⋮
FID0_ROIWH4 = FID1_ROIWH4 = FID2_ROIWH4 = FID3_ROIWH4

The register settings should satisfy following conditions:

- * Do not designate area like be overlap.

$ROI_{PH1} + ROI_{WH1} \leq ROI_{PH2}$

$ROI_{PH2} + ROI_{WH2} \leq ROI_{PH3}$

$ROI_{PH3} + ROI_{WH3} \leq ROI_{PH4}$

$ROI_{PH4} + ROI_{WH4} \leq$ The horizontal width at All-pixel scan.

$ROI_{PV1} + ROI_{WH1} \leq ROI_{PV2}$

$ROI_{PV2} + ROI_{WH2} \leq ROI_{PV3}$

$ROI_{PV3} + ROI_{WH3} \leq ROI_{PV4}$

$ROI_{PV4} + ROI_{WH4} \leq$ The vertical width at All-pixel scan.

- * Set the horizontal and vertical setting in multiple of 16.

(IMX530 series, IMX535 series)

- * Set the horizontal and vertical setting in multiple of 8.

(IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series)

- * Minimum width of the window is as below.

$ROI_{WH1} + ROI_{WH2} + ROI_{WH3} + ROI_{WH4} \geq 16d$

(IMX530 series, IMX535 series)

$ROI_{WH1} + ROI_{WH2} + ROI_{WH3} + ROI_{WH4} \geq 8d$

(IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series)

Reflection timing of FID is same as Multi Frame Set Output Mode.

5.2. In Case of Overlap ROI

The setting of ROI is basically similar to the datasheet. Refer to the datasheet for details.

The setting of Multi Frame concerns is basically similar to the application note. Refer to the application note of Multi Frame Set Output mode.

However, Multi Frame ROI have the following restrictions.

Set VMAX (Master mode) or the period of XVS (Slave mode) to the maximum value of each frame.

About frame line number, refer to the datasheet.

- ◆ Cropping position can set maximum 4 areas, regarding effective pixel start position as origin (0, 0) in all pixel scan mode.

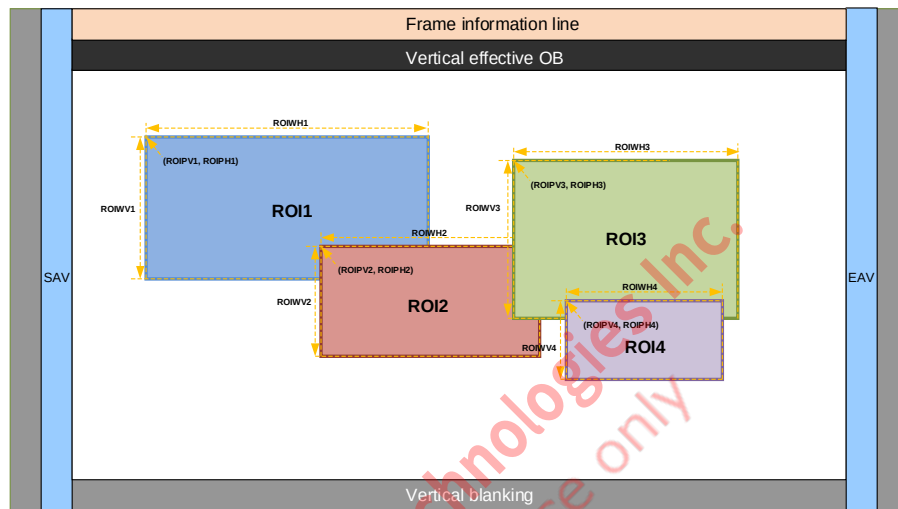
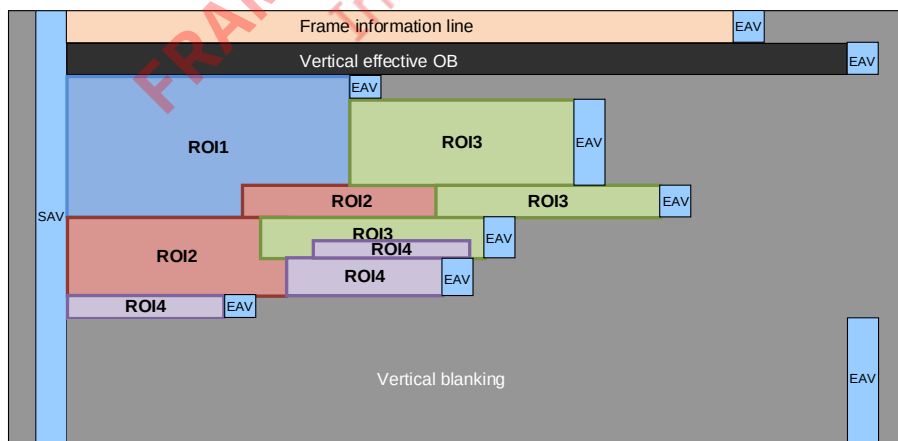


Image Drawing of Designated Areas in ROI Mode (SLVS)



Details of Image Drawing (SLVS)

- ◆ Multi Frame ROI is able to set cropping address and the values of ROI cropping size of each frame. But set the same number of ROI cropping windows of each frame.

$ROI_{PH1} + ROI_{WH1} \leq$ The horizontal width at All-pixel scan.

$ROI_{PH2} + ROI_{WH2} \leq$ The horizontal width at All-pixel scan.

$ROI_{PH3} + ROI_{WH3} \leq$ The horizontal width at All-pixel scan.

$ROI_{PH8} + ROI_{WH8} \leq$ The horizontal width at All-pixel scan.

$ROI_{PV1} + ROI_{WV1} \leq$ The vertical width at All-pixel scan.

$ROI_{PV2} + ROI_{WV2} \leq$ The vertical width at All-pixel scan.

$ROI_{PV3} + ROI_{WV3} \leq$ The vertical width at All-pixel scan.

$ROI_{PV8} + ROI_{WV8} \leq$ The vertical width at All-pixel scan.

* Set the horizontal and vertical setting in multiple of 16.

(IMX530 series, IMX535 series)

* Set the horizontal and vertical setting in multiple of 8.

(IMX540 series, IMX545 series, IMX548 series,
IMX565 series, IMX568 series)

* Minimum width of the window is as below.

$ROI_{WV1} \geq 16d$, $ROI_{WV2} \geq 16d$, $ROI_{WV3} \geq 16d$, $ROI_{WV4} \geq 16d$ (IMX530 series, IMX535 series)

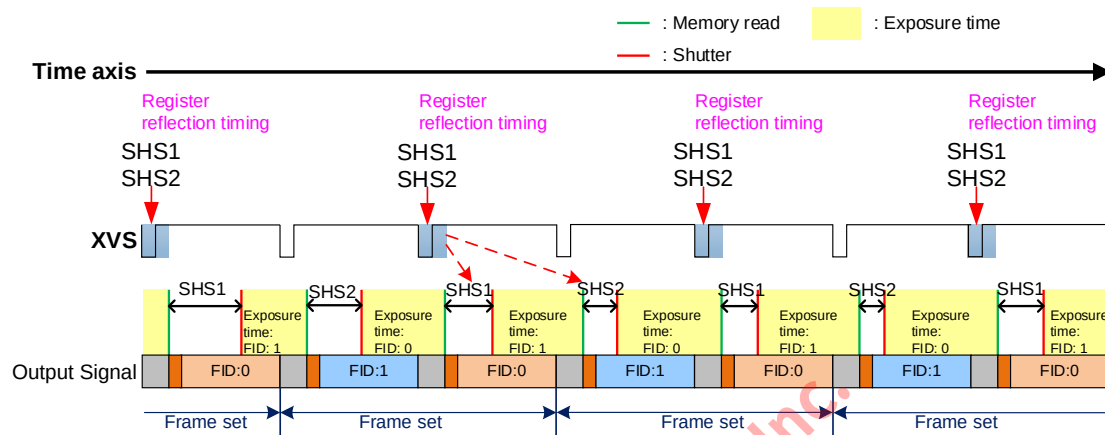
$ROI_{WV1} \geq 8d$, $ROI_{WV2} \geq 8d$, $ROI_{WV3} \geq 8d$, $ROI_{WV4} \geq 8d$ (IMX540 series, IMX545 series, IMX548 series,
IMX565 series, IMX568 series)

Reflection timing of FID is same as Multi Frame Set Output Mode.

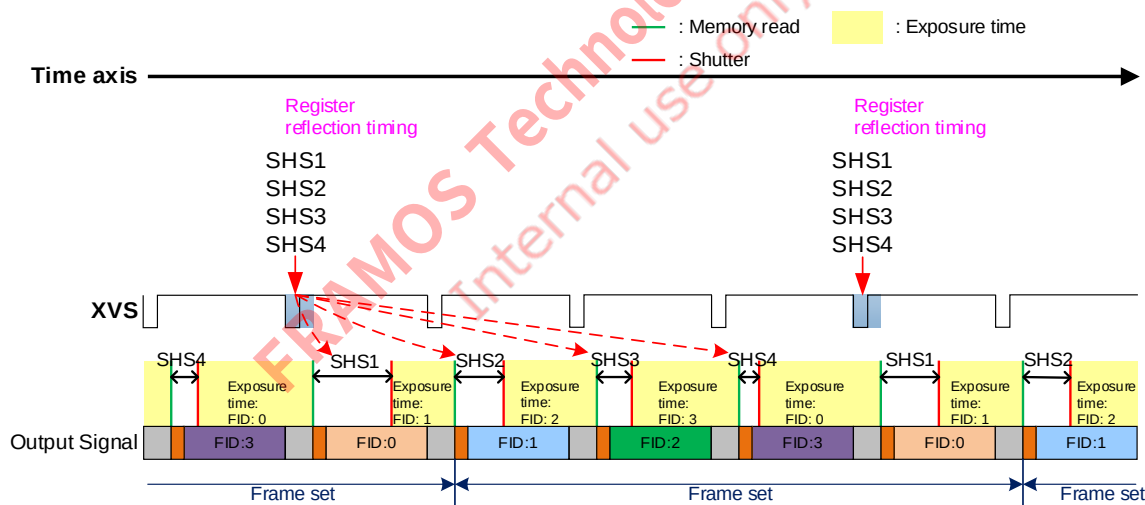
6. Reflection Timing for Exposure period

In Multi Frame ROI mode, the setting of register is reflected only if the Frame ID is "0d". (About the FID, refer to next page.) Set the shutter settings in such a value that the magnitude relation is satisfied.

- ◆ 2 frame set: SHS1 > SHS2
- ◆ 4 frame set: SHS1 > SHS2 > SHS3 > SHS4



Timing of register setting for Multi Frame ROI 2 frames



Timing of register setting for Multi Frame ROI 4 frames

The exposure time of each frame is according to value the following register.

Frame ID	Multi Frame ROI 2 frames [H]	Multi Frame ROI 4 frames [H]
0	The number of line of 1 frame - (SHS2) + toffset*	The number of line of 1 frame - (SHS4) + toffset*
1	The number of line of 1 frame - (SHS1) + toffset*	The number of line of 1 frame - (SHS1) + toffset*
2	—	The number of line of 1 frame - (SHS2) + toffset*
3	—	The number of line of 1 frame - (SHS3) + toffset*

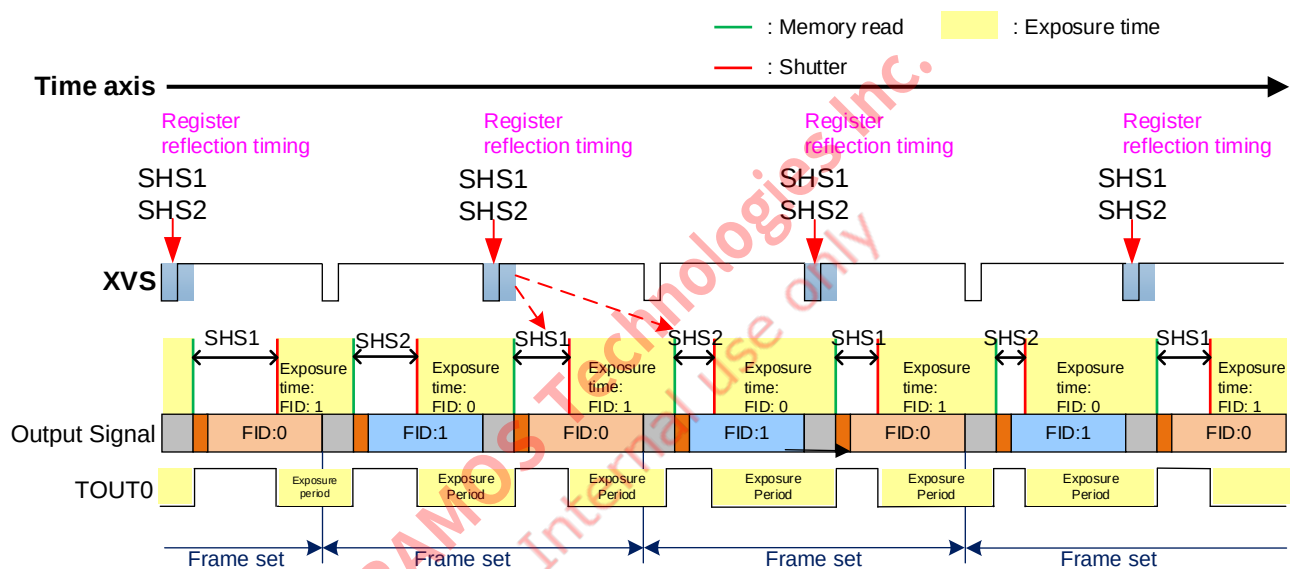
* Please refer to product specifications for toffset

7. The monitor output for Exposure period

As for this sensor, TOUT0 can output the status of the exposure control signal in Multi Frame ROI mode.

7.1. List of registers

Register	Register details			Initial value	Description
	Chip-ID	Address (): I2C	bit		
TOUT0SEL [2:0]	06h	36h (3436h)	[2:0]	0h	TOUT0 pin setting 0h : Low fixed 1h : Exposure period monitoring



Exposure period monitor for Multi Frame ROI 2 Frame

8. Each frame Gain mode

The sensor becomes Multi Frame Set Output mode that the exposure times of each frame are controlled by setting register WDSEL. Then, set to each frame is available by setting 1h to register WD_GAIN_MODE. Thereby the gain settings set each frame, analog and digital individually. Setting step is 0.1 dB.

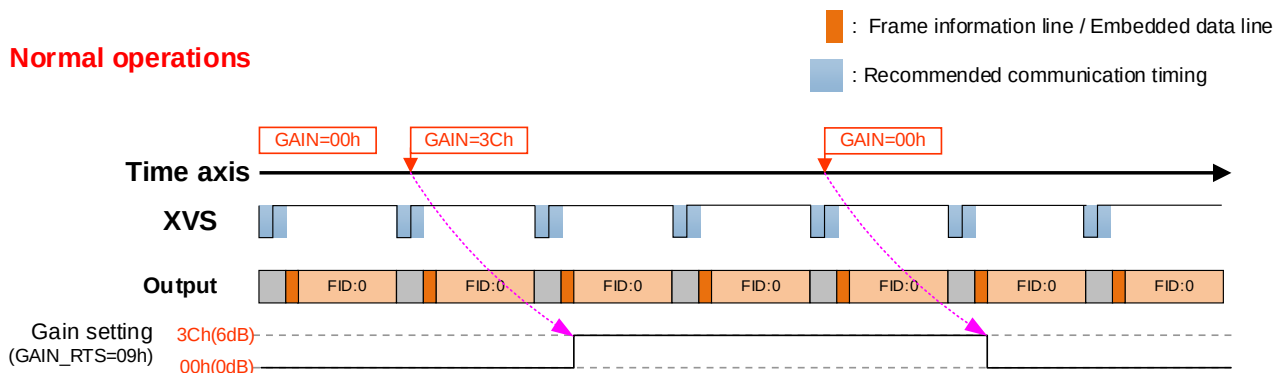
8.1. List of setting registers

Register details			Register	Description	Normal mode	Multi Frame Set Output mode	
Chip-ID	Address (): I2C	bit				2 frames	4 frames
04h	08h (3208h)	[1:0]	WDSEL [1:0]	In Multi Frame Set Output mode, set number of operation frames.	0h	1h	3h
07h	00h (3500h)	[6]	WD_GAIN_MODE [0]	Individually gain control	Fixed to "0h"	Disable: 0h Enable: 1h	Disable: 0h Enable: 1h
	14h (3514h)	[7:0]	GAIN [8:0]	Gain setting	Gain setting 0 to 48 dB	Frame ID: 0h Gain setting 0 to 48 dB	Frame ID: 0h Gain setting 0 to 48 dB
	15h (3515h)	[0]					
	16h (3516h)	[7:0]	GAIN1 [8:0]	Gain Setting	Fixed to 000h	In WD_GAIN_MODE = 1h, Frame ID: 1h Gain setting 0 to 48 dB	In WD_GAIN_MODE = 1h, Frame ID: 1h Gain setting 0 to 48 dB
	17h (3517h)	[0]					
	18h (3518h)	[7:0]	GAIN2 [8:0]	Gain setting	Fixed to 000h	Fixed to 000h	In WD_GAIN_MODE = 1h, Frame ID: 2h Gain setting 0 to 48 dB
	19h (3519h)	[0]					
	1Ah (351Ah)	[7:0]	GAIN3 [8:0]	Gain Setting	Fixed to 000h	Fixed to 000h	In WD_GAIN_MODE = 1h, Frame ID: 3h Gain setting 0 to 48 dB
	1Bh (351Bh)	[0]					

9. Gain reflection timing

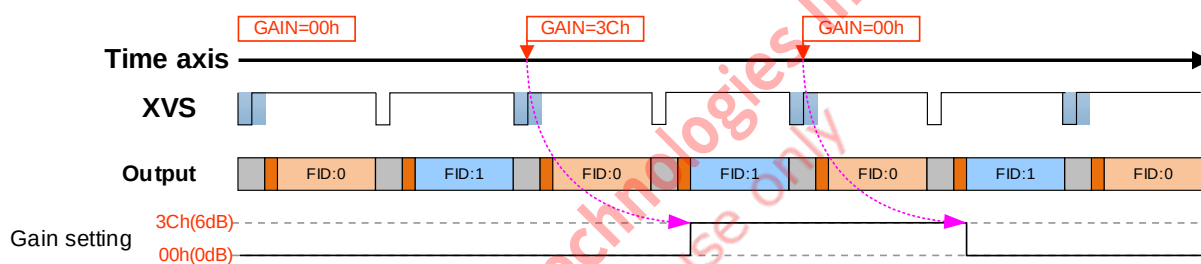
In Multi Frame ROI mode, the reflection timing of gain setting is reflected collectively to the unit of frameset. The gain reflection timing is shown figure below. (In Multi Frame ROI mode, a special setting is unnecessary.)

Normal operations



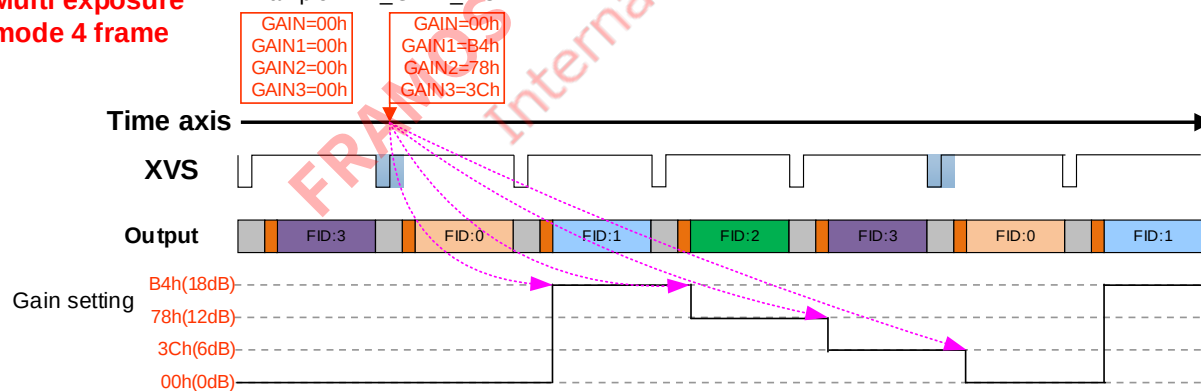
Multi exposure mode 2 frame

Example: WD_GAIN_MODE = 0



Multi exposure mode 4 frame

Example: WD GAIN MODE = 1



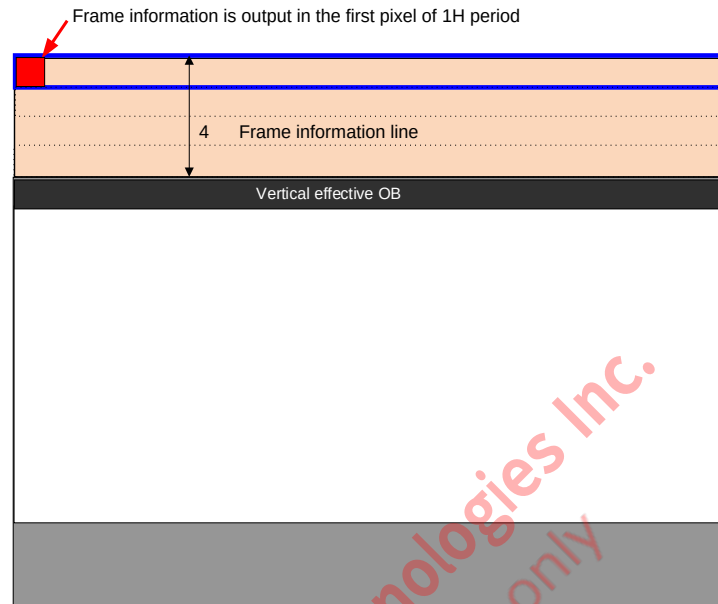
Gain Reflection Timing

10. Frame identification method

The identification signal of each frame is Frame ID, and the unit of Multi Frame Set Output mode is Frame set.

10.1. Frame information line

This product uses Frame information line of SLVS to output Frame ID and Frame set. It is added to the output signal of the 1st pixel of the blue frame in the figure below.



Frame information line output position

10.2. Output format (Frame information line)

The output format of Frame information line is shown below.

Output	8bit
[7]	1
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

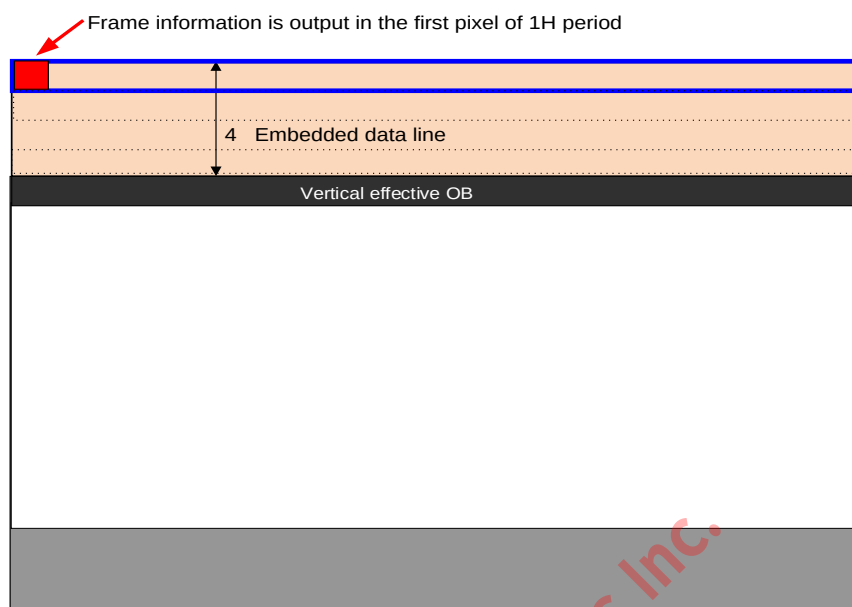
Output	10bit
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	12bit
[11]	0
[10]	0
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output format (Frame information line)

10.3. Embedded data line

This product uses Embedded data line of SLVS – EC or CSI-2 to output Frame ID and Frame set. It is added to the output signal of the 1st pixel of the blue frame in the figure below.



Embedded data line output position

10.4. Output format (Embedded data line)

The output format of embedded data line is shown below.

(SLVS-EC)

Output	8bit
[7]	1
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	12bit
[11]	0
[10]	0
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output format (Embedded data line (SLVS-EC))

(CSI-2)

Output	8bit
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	Frame ID [1]
[8]	Frame ID [0]
[7]	0
[6]	0
[5]	0
[4]	Frameset
[3]	Frame ID [1]
[2]	Frame ID [0]
[1]	0
[0]	1

Output	12bit
[11]	Frame ID [1]
[10]	Frame ID [0]
[9]	0
[8]	0
[7]	0
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	1
[1]	0
[0]	1

Output format (Embedded data line (CSI-2))

11. Mode Transitions

The transition related to Multi Frame ROI mode is shown below. If perform the transition which is not described in table below, set via sensor standby.

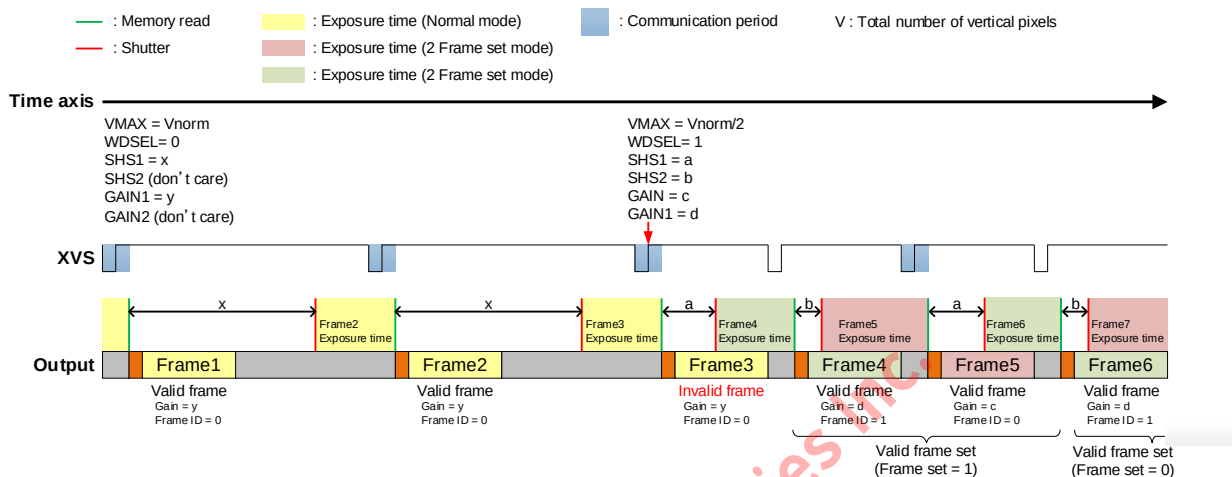
11.1. Mode Transition

Mode Transition		Remarks
Normal mode (ROI mode)	⇔ Multi Frame ROI (2 Frame)	There comes out invalid frames. Changing the register value of ROI cropping position and cropping size is prohibited. Changing the register value of FREQ, HMAX is prohibited. In case of setting as the same frame rate, set the 1 frame period of normal mode which to be same as 1 frame set period of Multi Frame Set Output mode.
Normal mode (ROI mode)	⇔ Multi Frame ROI (4 Frame)	There comes out invalid frames. Changing the register value of ROI cropping position and cropping size is prohibited. Changing the register value of FREQ, HMAX is prohibited. In case of setting as the same frame rate, set the 1 frame period of normal mode which to be same as 1 frame set period of Multi Frame Set Output mode.
Multi Frame ROI (2 Frame)	⇔ Multi Frame ROI (4 Frame)	Set via the sensor standby.
Change ROI cropping position and cropping size		Changing the ROI cropping position and cropping size is prohibited even within the same frame set drive. Set via the sensor standby.
Multi Frame ROI (2 Frame)	⇔ Multi Frame ROI (2 Frame)	
Multi Frame ROI (4 Frame)	⇔ Multi Frame ROI (4 Frame)	

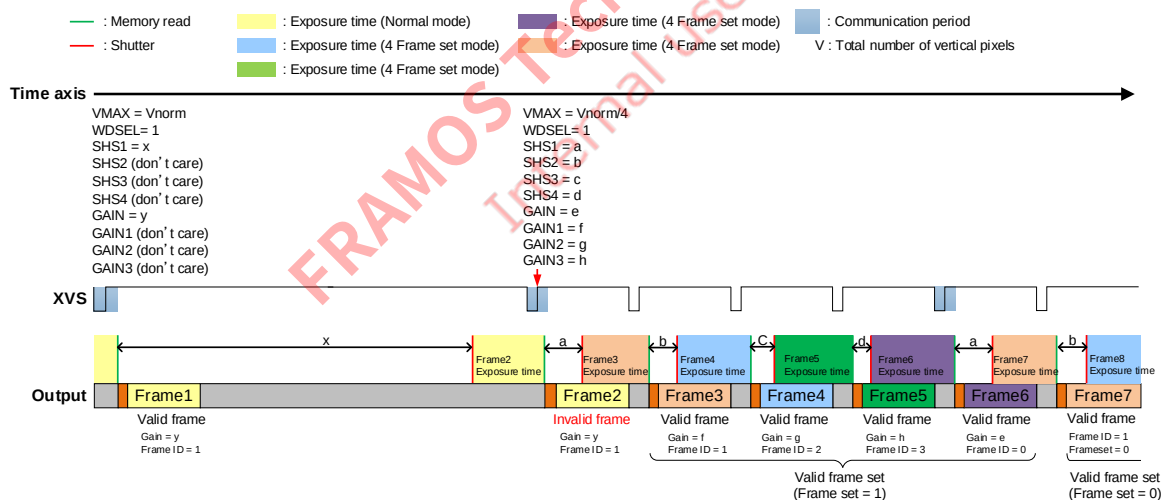
As one example, in case of setting to the same frame rate before the mode transition and after the mode transition, it is shown as follows.

◆ Transition from normal ROI mode to Multi Frame ROI mode

The transition setting frame becomes invalid.



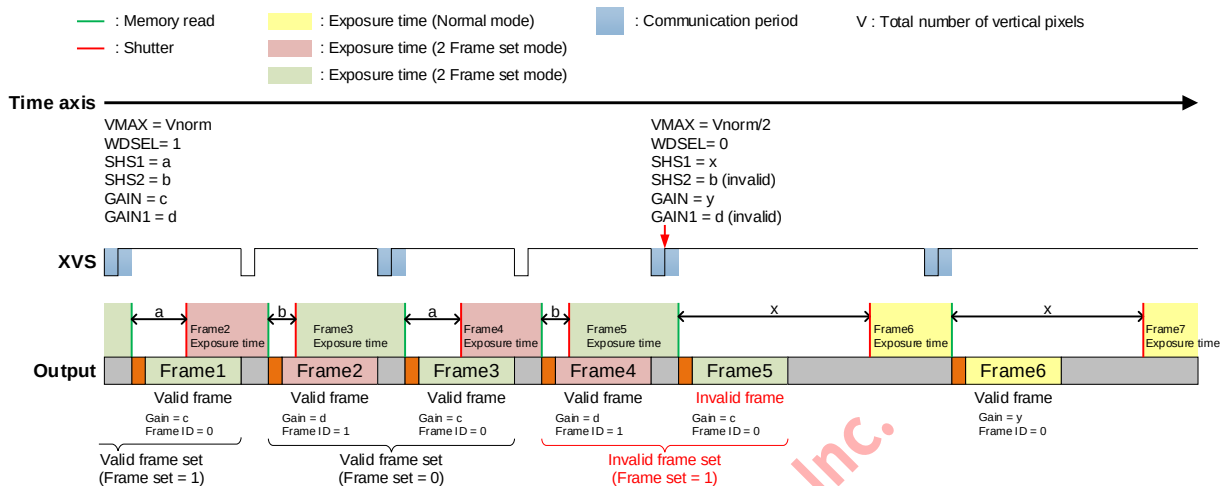
Transition to Multi Frame ROI mode (2 frames) from Normal ROI mode (Each frame gain mode)



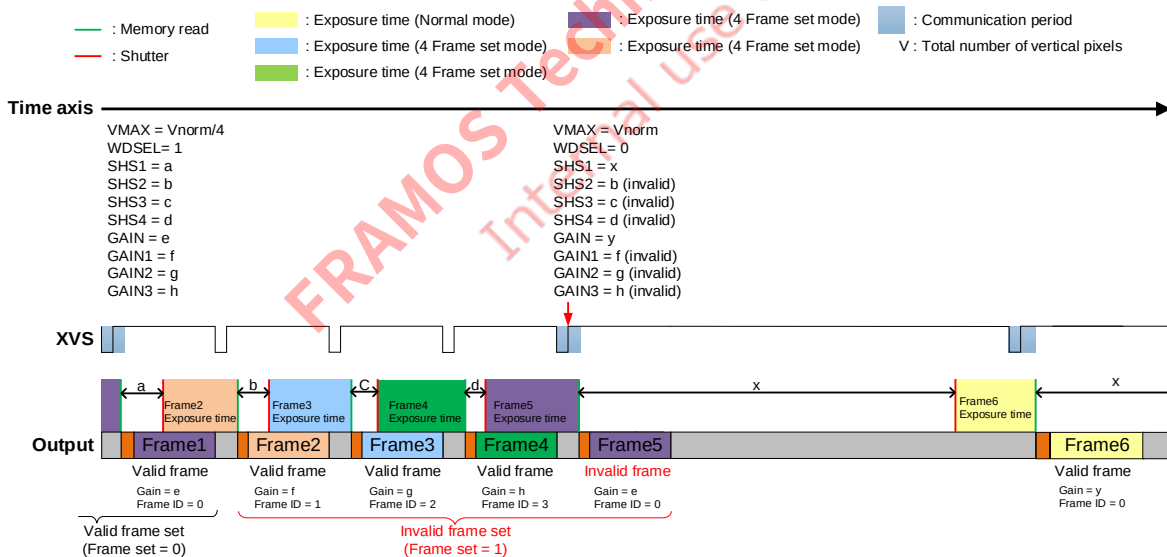
Transition to Multi Frame ROI mode (4 frames) from Normal ROI mode (Each frame gain mode)

◆ Transition from Multi Frame ROI mode to normal ROI mode

The transition setting frame becomes invalid. Because the transition setting frame is long term exposure signal of Multi Frame ROI mode frame set, the last signal set become invalid.



Transition to Normal ROI mode from Multi Frame ROI mode (2 frames) (Each frame gain mode)



Transition to Normal ROI mode from Multi Frame ROI mode (4 frames) (Each frame gain mode)

12. Notes On Handling

Notes On Handling of the Multi Frame ROI mode.

- ◆ Multi Frame ROI mode and Dual ADC (without combination / with built-in combination) come by combination drive.
(only IMX530 series and IMX535series)
- ◆ The drive with the trigger mode is non-support.
- ◆ Multi Frame ROI mode and Short interval shutter function do not come by combination drive.
- ◆ Multi Frame ROI mode and Self Trigger do not come by combination drive.

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Internal use only

Revision History

Version	Date	Page	Remarks
Rev .0.1	2019 / 4 / 26	-	First Edition
Rev 0.2	2019 / 6 / 28	1, 2, 37, 38	Correction : Function name Exposure time ⇒ Exposure period
		46	Correction : Function name Dual ADC (without combination / with built-in combination) Short interval shutter function
Rev 1.0	2020 / 6 / 4	–	First Edition (Official version of IMX530 series / IMX540 series
		1	Added Chapter “Product type”
		2	Added SLVS 2 ch output in support interface Added the restriction in support interface
		35, 37	Update the limitation of ROI
		41	Update the Figure of Gain Reflection Timing
Rev.2.0	2020 / 12 / 17	–	Added: IMX535 series / IMX545 series / IMX565 series
		2, 44	Added: CSI-2 serial data output
		28, 29, 30	Added: Default Value after reset of IMX535 series, IMX545 series, IMX565 series
		6 to 27 35, 37	Added: Registers value of IMX535 series, IMX545 series, IMX565 series
Rev.3.0	2021 / 3 / 4	1, 2, 27, 30, 35, 37	Added: IMX548 series / IMX568 series
		35	Correction: ROI specified setting conditions "≤" ⇒ "≤"
		48	Added: Definition of series for Dual ADC combination drive